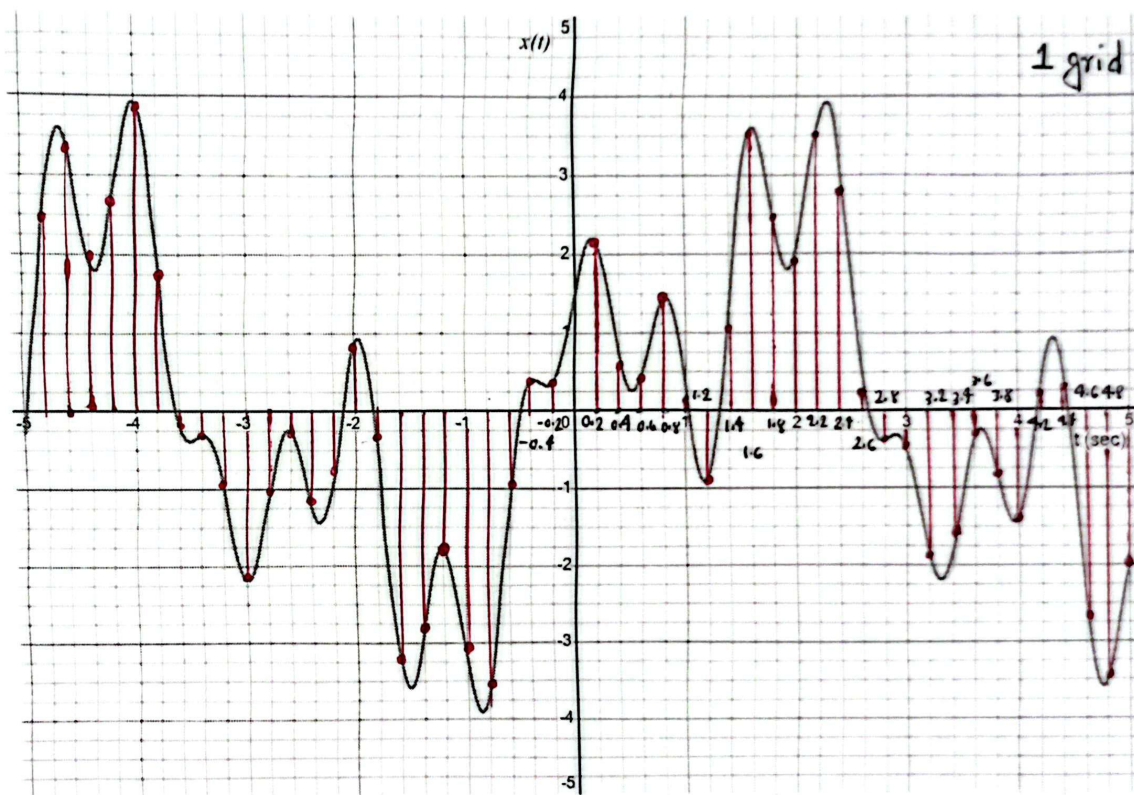




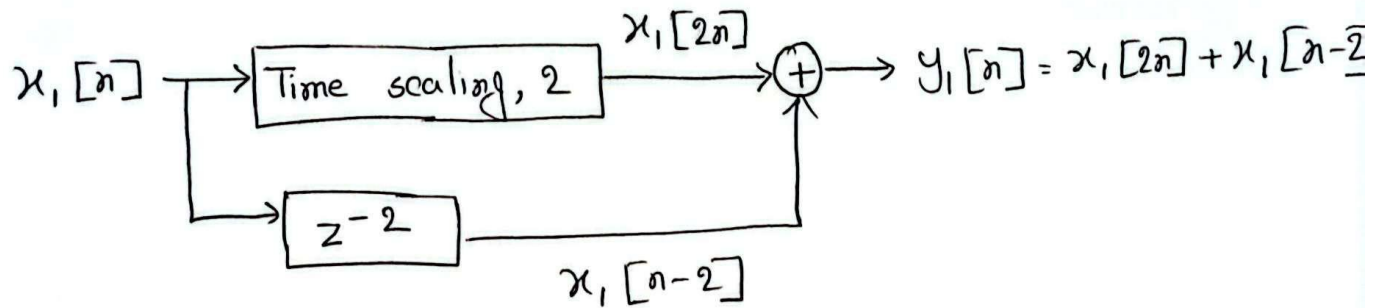
1 grid = 0.5 unit

$f_s = 5 \text{ Hz}$

$\therefore T_s = 0.2 \text{ s}$

Ans-2

(1)

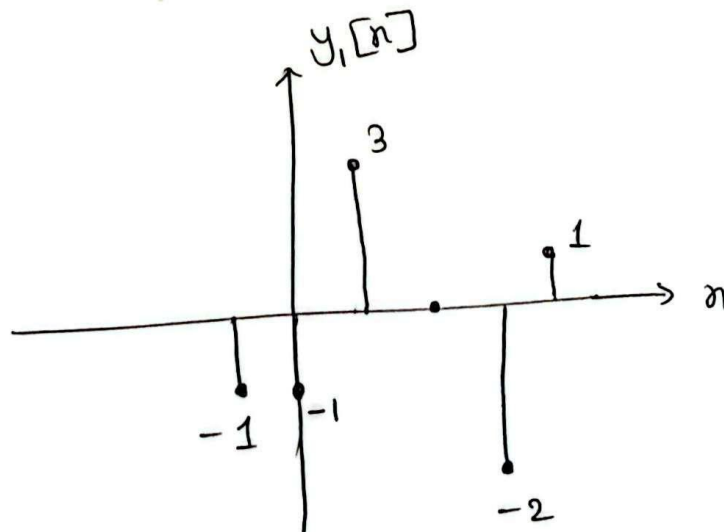


$$x_1[n] = \{-1, 2, \underset{\uparrow}{0}, -2, 1\}$$

$$y_1[n] = x_1[2n] + x_1[n-2]$$

$$= \{-1, \underset{\uparrow}{0}, 1\} + \{-1, 2, \underset{\uparrow}{0}, -2, 1\}$$

$$= \{-1, \underset{\uparrow}{-1}, 3, 0, -2, 1\}$$



Block diagram of a discrete-time system:

Input $x_2[n]$ splits into two paths:

- Path 1: Through a delay block z^{-3} to produce $x_2[n+3]$.
- Path 2: Directly to a subtraction node $-$.

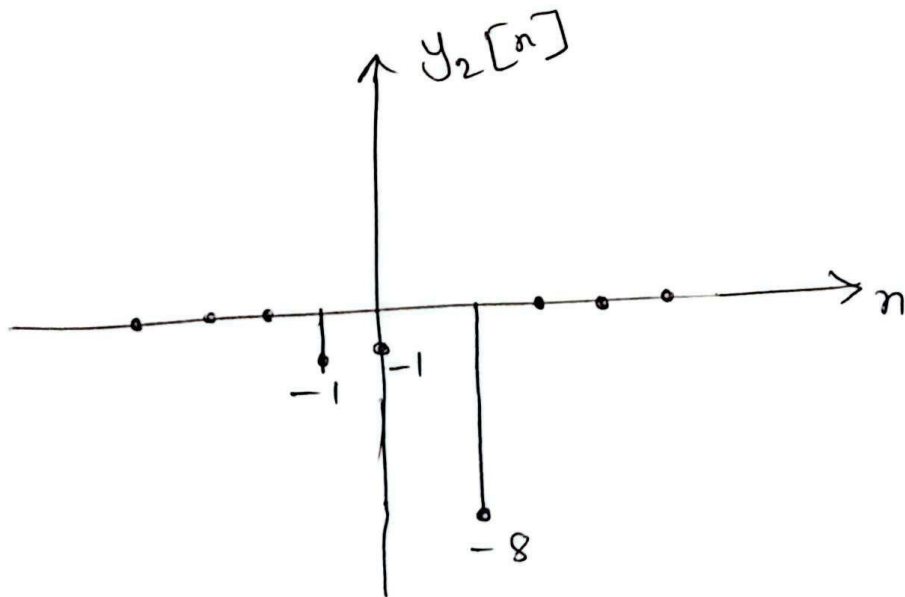
The subtraction node also receives a constant input of 4. The output of the subtraction node is multiplied by $x_2[n+3]$ at a multiplication node $\times$ to produce the final output $y_2[n]$.

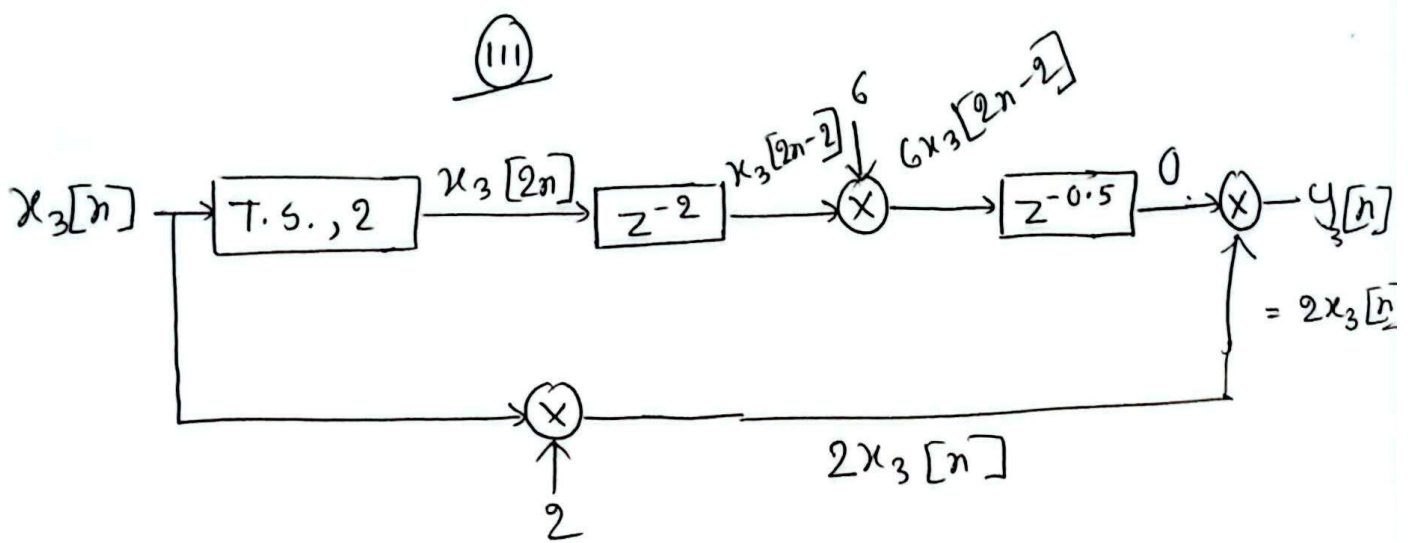
Equation for the output:

$$y_2[n] = x_2[n+3] \times (x_2[n] - 4)$$

$$y_2[n] = x_2[n+3] \times x_2[n] - 4$$

$$= \{0, 0, 0, -1, \underset{\uparrow}{-1}, -8, 0, 0, 0\}$$





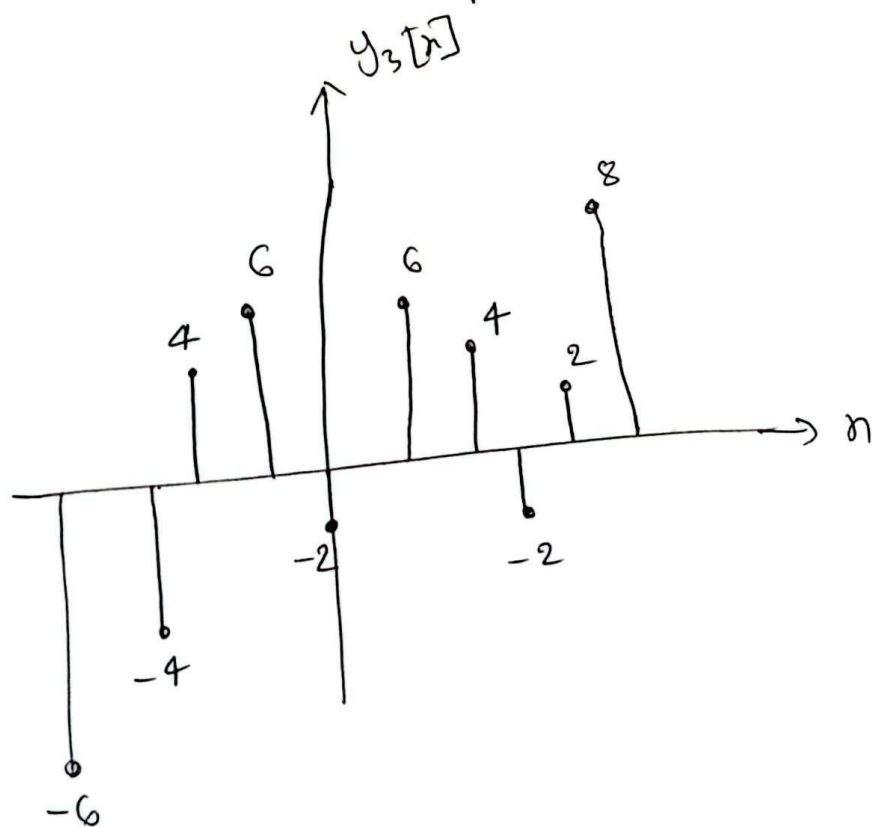
$$x_3[n] = \{-3, -2, 2, 3, -1, 3, 2, -1, 1, 4\}$$

↑

$$y_3[n] = \{-6, -4, 4, 6, -2, 6, 4, -2, 2, 8\}$$

$$= \{-6, -4, 4, 6, -2, 6, 4, -2, 2, 8\}$$

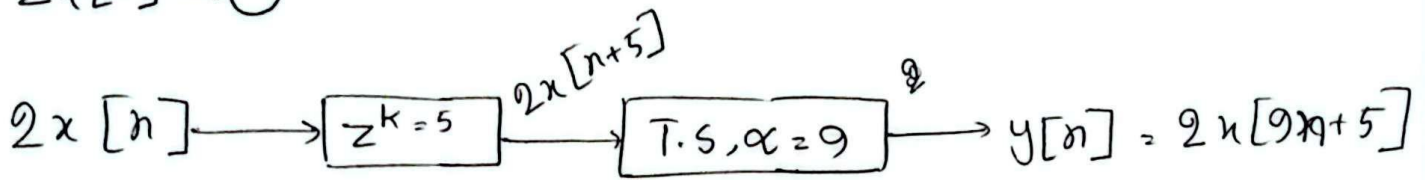
↑



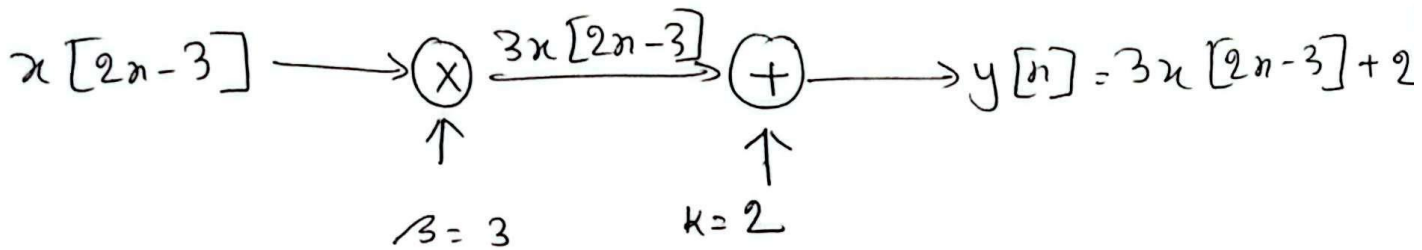
Ans-3

①

~~$2x[n] \rightarrow \text{circle}$~~



②



③

